



8961 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	B10	NIRSpec MultiObject Spectroscopy	(1) B10
	2	B10-background	NIRSpec MultiObject Spectroscopy	(5) B10-background-region
	3	B211	NIRSpec MultiObject Spectroscopy	(2) B211
	4	B211-background	NIRSpec MultiObject Spectroscopy	(6) B211_background-region
	5	B218	NIRSpec MultiObject Spectroscopy	(3) B218
	6	B218-slit2	NIRSpec MultiObject Spectroscopy	(4) B218_slit2
	7	B218-background	NIRSpec MultiObject Spectroscopy	(7) B218-background-region

ABSTRACT

Cosmic rays (CRs) significantly shape the thermal and chemical state of dense molecular gas, yet measuring the CR ionization rate (CRIR)—a key parameter in astrochemical models—remains challenging. Traditional methods rely on rare ion absorption measurements toward bright background

sources or complex model fitting to emission lines. Recently, the “H₂ direct method” was developed, utilizing CR-induced H₂ rovibrational emission to measure the CRIR without the need for background stars or model fitting. In a pioneering JWST observation of Barnard 68 (GO 5064), CR-induced H₂ lines were detected for the first time, enabling a direct measurement of the CRIR in the cloud, providing strong proof-of-concept for this method.

Building on this success, we propose a survey to measure the CRIR in three dense, starless cores within the Taurus molecular cloud complex. We will also observe the envelope of one core to investigate CRIR variations on sub-parsec scales (<0.1 pc). This will yield CRIR measurements across a range of scales, from cloud core scales to ~ 10 pc and up to ~ 300 pc (when complemented with Barnard 68 observations), allowing us to study CR propagation in the interstellar medium and dense cloud cores. The CRIR is a significant uncertainty in understanding CR interactions with molecular clouds. This proposal will deliver robust CRIR measurements across a well-characterized star-forming region and be the first to probe sub-parsec scale CRIR variations within a molecular cloud, offering new insights into the CRIR distribution, the sources of low-energy CRs, and their propagation.

OBSERVING DESCRIPTION

We propose to observe the starless core, B68, using the MOS-NIRSpec instrument.

Since the angular size of B68 is well matched to the field of view of NIRSpec, we plan to use the MOS mode to collect light from a large portion of the cloud. In practice we use the MSA to compose a long slit, and then we will stack the spectra from all (or a significant part of the) shutters to obtain a deep spectrum.

We adopt the G235H grating because it covers the most interesting spectral region in the near-infrared range, where many CR-sensitive emission lines are found. We choose the high-resolution grating, which yields a minimal loss in SNR compared to the medium-resolution grating, because it allows us to resolve the 1-0 Q(1) line from the 1-0 Q(2) line.

Proposal 8961 - Targets - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	B10	RA: 04 17 39.6417 (64.4151738d) Dec: +28 03 49.15 (28.06365d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas]</i> <i>Extended=YES</i>		
	(2)	B211	RA: 04 18 7.8712 (64.5327967d) Dec: +27 34 12.55 (27.57015d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas]</i> <i>Extended=YES</i>		
	(3)	B218	RA: 04 27 48.9411 (66.9539213d) Dec: +26 18 3.68 (26.30102d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas]</i> <i>Extended=YES</i>		
	(4)	B218_slit2	RA: 04 27 44.5522 (66.9356342d) Dec: +26 19 3.17 (26.31755d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas]</i> <i>Extended=YES</i>		
	(5)	B10-background-region	RA: 04 18 39.5970 (64.6649875d) Dec: +28 03 4.41 (28.05123d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular clouds]</i> <i>Extended=YES</i>		
	(6)	B211_background-region	RA: 04 18 45.9669 (64.6915287d) Dec: +27 32 21.47 (27.53930d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas]</i> <i>Extended=YES</i>		
	(7)	B218-background-region	RA: 04 28 7.0216 (67.0292567d) Dec: +26 12 23.60 (26.20656d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Molecular clouds]</i> <i>Extended=YES</i>		

Proposal 8961 - Observation 1 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 1: B10										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observations:[B10-background (Obs 2)]										
Diagnostics	(B10 (Obs 1)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	B10	RA: 04 17 39.6417 (64.4151738d)								
			Dec: +28 03 49.15 (28.06365d)								
			Equinox: J2000								
	Comments: Category=ISM Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas] Extended=YES										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.9312662976110 437E-4	-10.0	-10.0	1	4	5893.912
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			1.9653053907052 4E-4	-5.0	-5.0	1	4	5893.912
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.999999999999 994	0.0	0.0	1	4	5893.912
	4	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99980353423 11	5.0	5.0	1	4	5893.912
	5	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99960713244 15	10.0	10.0	1	4	5893.912
Special Requirements	Aperture PA Range 230 to 238 Degrees (V3 91.42422485 to 99.42422485)										
	Group Observations 1, 2, Non-interruptible										

Proposal 8961 - Observation 2 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 2: B10-background										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observation For: [B10 (Obs 1)]										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(5)	B10-background-region	RA: 04 18 39.5970 (64.6649875d)								
			Dec: +28 03 4.41 (28.05123d)								
			Equinox: J2000								
	Comments: Category=ISM Description=[Molecular clouds] Extended=YES										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.9291957635866 634E-4	-10.0	-10.0	1	1	1473.478
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.999999999999 994	0.0	0.0	1	1	1473.478
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99980363769 725	5.0	5.0	1	1	1473.478
Special Requirements	Group Observations 1, 2, Non-interruptible										

Proposal 8961 - Observation 3 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 3: B211										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observations:[B211-background (Obs 4)]										
Diagnostics	(B211 (Obs 3)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	B211	RA: 04 18 7.8712 (64.5327967d) Dec: +27 34 12.55 (27.57015d) Equinox: J2000								
	Comments: Category=ISM Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas] Extended=YES										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.8494109505831 097E-4	-10.0	-10.0	1	4	5893.912
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			1.9243857467937 013E-4	-5.0	-5.0	1	4	5893.912
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99999999999 994	0.0	0.0	1	4	5893.912
	4	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99980762460 62	5.0	5.0	1	4	5893.912
	5	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99961531161 9	10.0	10.0	1	4	5893.912
Special Requirements	Aperture PA Range 224 to 230 Degrees (V3 85.42422485 to 91.42422485)										
	Group Observations 3, 4, Non-interruptible										

Proposal 8961 - Observation 4 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 4: B211-background										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observation For: [B211 (Obs 3)]										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(6)	B211_background-region	RA: 04 18 45.9669 (64.6915287d) Dec: +27 32 21.47 (27.53930d) Equinox: J2000								
	Comments: Category= <i>ISM</i> Description=[<i>Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas</i>] Extended= <i>YES</i>										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.8443176092036 3E-4	-10.0	-10.0	1	1	1473.478
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.999999999999 994	0.0	0.0	1	1	1473.478
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99980787912 45	5.0	5.0	1	1	1473.478
Special Requirements	Group Observations 3, 4, Non-interruptible										

Proposal 8961 - Observation 5 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 5: B218										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observations:[]										
Diagnostics	(B218 (Obs 5)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	B218	RA: 04 27 48.9411 (66.9539213d)								
			Dec: +26 18 3.68 (26.30102d)								
			Equinox: J2000								
	Comments: Category=ISM Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas] Extended=YES										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.6422185782059 263E-4	-10.0	-10.0	1	4	5893.912
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			1.8208098333702 397E-4	-5.0	-5.0	1	4	5893.912
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			0.0	0.0	0.0	1	4	5893.912
	4	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99981797818 486	5.0	5.0	1	4	5893.912
	5	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99963601480 613	10.0	10.0	1	4	5893.912
Special Requirements	Aperture PA Range 224.6 to 245 Degrees (V3 86.02422485 to 106.42422485)										

Proposal 8961 - Observation 6 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 6: B218-slit2											Fri Feb 13 14:00:13 GMT 2026		
	Diagnostic Status: Warning													
	Observing Template: NIRSpec MultiObject Spectroscopy													
	Background Observations:[]													
Diagnostics	(B218-slit2 (Obs 6)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure													
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates		Targ. Coord. Corrections				Miscellaneous				
	(4)	B218_slit2		RA: 04 27 44.5522 (66.9356342d)										
				Dec: +26 19 3.17 (26.31755d)										
				Equinox: J2000										
	Comments: Category=ISM Description=[Compact nebulae, Dark interstellar clouds, Dust nebulae, Molecular clouds, Molecular gas] Extended=YES													
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold		
	NONE	false		No	Q4 Field Point 1					jwst-nirspec-g235h		1.5		
Spectral Elements	#	Exposure Specification		MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time		
	1	1 (G235H/F170LP)		Q4 Field Point 1 Long Slit			3.6448867750488 31E-4	-10.0	-10.0	1	4	5893.912		
	2	1 (G235H/F170LP)		Q4 Field Point 1 Long Slit			1.8221436712104 864E-4	-5.0	-5.0	1	4	5893.912		
	3	1 (G235H/F170LP)		Q4 Field Point 1 Long Slit			359.999999999999 994	0.0	0.0	1	4	5893.912		
	4	1 (G235H/F170LP)		Q4 Field Point 1 Long Slit			359.99981784485 27	5.0	5.0	1	4	5893.912		
	5	1 (G235H/F170LP)		Q4 Field Point 1 Long Slit			359.99963574819 27	10.0	10.0	1	4	5893.912		
Special Requirements	Aperture PA Range 224.6 to 235 Degrees (V3 86.02422485 to 96.42422485)													

Proposal 8961 - Observation 7 - Probing the cosmic ray ionization rate with H2 rovibrational line emission - the Taurus survey

Observation	Proposal 8961, Observation 7: B218-background										Fri Feb 13 14:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Background Observation For: []										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(7)	B218-background-region	RA: 04 28 7.0216 (67.0292567d) Dec: +26 12 23.60 (26.20656d) Equinox: J2000								
	Comments: Category=ISM Description=[Molecular clouds] Extended=YES										
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	NONE	false	No	Q4 Field Point 1			jwst-nirspec-g235h	1.5			
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			3.6269801676483 713E-4	-10.0	-10.0	1	1	1473.478
	2	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.999999999999 994	0.0	0.0	1	1	1473.478
	3	1 (G235H/F170LP)	Q4 Field Point 1 Long Slit			359.99981873966 203	5.0	5.0	1	1	1473.478